

Louis Pasteur

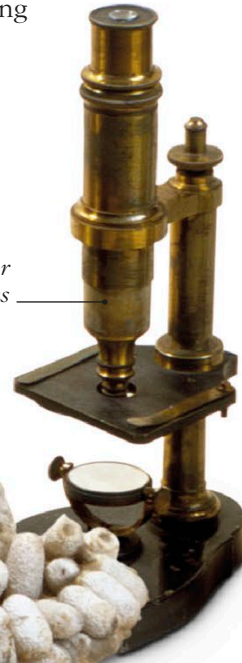
French biologist and chemist Louis Pasteur was one of the giant figures of 19th-century medicine. His many discoveries, particularly in the fields of germ theory and vaccination (see pp.242–243), revolutionized the scientific thinking of the day. They also saved millions of lives by providing great leaps forward in the treatment of deadly diseases, such as anthrax and rabies.



Workers examine the pasteurization tanks in a milk treatment depot in London, 1935

PASTEURIZATION

In the 1860s, Pasteur developed a way of preventing liquids, such as wine and milk, from spoiling because of contamination by bacteria. His technique involves steadily heating the liquid, which kills the bacteria without changing the liquid's taste. This process is known as pasteurization.



Microscope that Pasteur used to study silkworms

THE SILK INDUSTRY

Pasteur saved France's silk industry when it was being ravaged by a silkworm disease. He discovered that the disease could be passed from adult silkworms to the young, so he recommended destroying the eggs of infected moths. This made sure that only the healthy silkworms survived and the disease was wiped out.



Silkworm cocoon from which silk is made

LIFE STORY

1822

Born on December 27 in Dole, France, Pasteur grows up to study chemistry. In 1848, he becomes a professor at the University of Strasbourg, France.

1859

He uses special sealed flasks (right) to demonstrate that food rots because of microbes in the air.

1860s

Having invented pasteurization in 1863, Pasteur suffers a stroke in 1868 that partially paralyzes his left side. He recovers over time and is able to continue his work.

